1 Bayes Correlated Equilibrium

Suppose there are players 1,2,...,I and a finite set of states Θ.

A **basic game** $G = \big((A_i, u_i)_{i=1}^I, \psi\big)$ consists of (i) actions A_i and utilities u_i, and (ii) a common prior with full support over the states, $\psi \in \Delta_{++}(\Theta)$[1].

An **information structure** $S = \big((T_i)_{i=1}^I, \pi\big)$ consists of (i) signals/types T_i aggregated as $T = T_1 \times ... \times T_I$, and (ii) type distribution $\pi : \Theta \to \Delta(T)$ a probability mapping of states to types.

Together, a basic game G and an information structure S define a standard incomplete information game (G,S).

Within an **incomplete information game** (G,S), a **decision rule** σ is the mapping $\sigma : T \times \Theta \to \Delta(A)$. A mediator observes realizations t,θ and privately suggests actions to each player. For players to follow these suggestions, we require an **'obedience'** condition.

Definition 1.1 (Obedience)**.** *Decision rule σ is obedient for the incomplete information game (G,S) if for all players i, all types t_i, and all actions a_i, we have*

$$\sum_{a_{-i},t_{-i},\theta} \psi(\theta) \cdot \pi((t_i,t_{-i})|\theta) \cdot \sigma((a_i, a_{-i})|(t_i,t_{-i}), \theta) \cdot u_i((a_i, a_{-i}), \theta)$$
$$\geq \sum_{a_{-i},t_{-i},\theta} \psi(\theta) \cdot \pi((t_i,t_{-i})|\theta) \cdot \sigma((a_i, a_{-i})|(t_i,t_{-i}), \theta) \cdot u_i((a_i', a_{-i}), \theta)$$

for all other actions $a_i' \in A_i$.

Then, the Bayes correlated equilibrium just requires obedience.

Definition 1.2 (Bayes Correlated Equilibrium)**.** *A decision rule σ is a Bayes correlated equilibrium (BCE) of (G,S) if it is obedient for (G,S).*

Note that this definition is a generalization of correlated equilibrium in the sense that we allow a finite set of states Θ, rather than fixing the state as in the original correlated equilibrium by Aumann (1987). Hence, if Θ is a singleton and we remove the conditioning on θ's and the mapping ψ, we reduce to a correlated equilibrium.

[1] The Δ_{++} notation means that the distribution of Θ has full support, i.e. $\psi(\theta) > 0$ for all θ.